IE 300 Case Study 3:
Heavy-Tailed Distributions and Reinsurance

The purpose of this case study is to give a brief introduction to a heavy-tailed distribution and its distinct behaviors in contrast with familiar light-tailed distributions in standard texts. You will learn about QQ plot, which is a popular tool for checking goodness-of-fit for a particular statistical model. You will also work on a real-life application of heavy-tailed distributions in reinsurance rate-making.

Reinsurance is a very important component of the global financial market. It allows insurers to take on risks that they would otherwise not be able to. Did you know that NASA buys insurance contracts for every rocket it launches and every satellite and probe it sends to the outer space? These equipments are so expensive that typical insurers would not be able to cover on their own. Therefore, they can go to the reinsurance market, slide up the coverage and transfer partial coverage to reinsurers that exceed their financial capabilities. By the end of this case study, you will learn basic principles of pricing a reinsurance contract.

Learning objectives:

1. Visualize the concepts in the Central Limit Theorem.
2. Identify cases where the Central Limit Theorem does not apply.
3. Reinforce the concept of cumulative distribution function.
4. Understand why and how QQ plot works for the assessment of goodness-of-fit.
5. Reinforce the concepts of conditional probability and conditional expectation.
6. Apply basic integration technique to compute mean excess function.
7. Learn about behaviors of a heavy-tailed distribution.
8. Learn how to use order statistics to estimate quantiles and mean excess function.
9. Develop intuition behind point estimators.

1 Central Limit Theorem

This section is to provide visualization of central limit theorem which you should already be familiar with. We provide examples for both discrete random variables and continuous random variables.

1.1 Bernoulli Random Variables

Suppose we are given a potentially biased coin, i.e., a coin that may not have an equal chance of landing on a head or a tail. We would like to estimate the probability of getting a head.

We can do so by tossing the coin many times and counting the number of heads. Suppose we toss the coin n times. For every i from 1 to n, let X_i be a random variable: 1 if the i^{th} coin shows heads and 0 otherwise. Let p the probability of a head and $q = 1 - p$ be the probability of a tail. Then $X_i \sim \text{Bernoulli}(p)$, i.e., X_i is a Bernoulli random variable with parameter p. Its probability mass function is given by

$$\mathrm{P}(X_i = x) = \begin{cases} p & \text{if } x = 1 \\ q & \text{if } x = 0 \end{cases}.$$

The total number of heads in n tosses is given by

$$S_n := \sum_{i=1}^{n} X_i.$$

Then $S_n \sim \text{Binomial}(n, p)$, i.e., S_n is a binomial random variable with parameters n and p. Its probability mass function is given by

$$\mathrm{P}(S_n = x) = \binom{n}{x} p^x q^{n-x}, \qquad x \in \{0, 1, \ldots, n\}.$$

The random variable

$$\overline{X}_n := \frac{1}{n} S_n$$

is called the *sample average*. What are the mean and variance of $\overline{X}_n$? To find them, we will use the following result.

Theorem 1.1. *For any real numbers a and b, and random variable Z, we have*

$$\mathbb{E}(aZ + b) = a\,\mathbb{E}(Z) \qquad\qquad \mathrm{Var}(aZ + b) = a^2\,\mathrm{Var}(Z)$$

Using Theorem 1.1, we get

$$\mathbb{E}(\overline{X}_n) = \mathbb{E}(S_n)/n = p, \quad \text{and}$$

$$\mathrm{Var}(\overline{X}_n) = \mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{\mathrm{Var}(S_n)}{n^2} = \frac{pq}{n}.$$

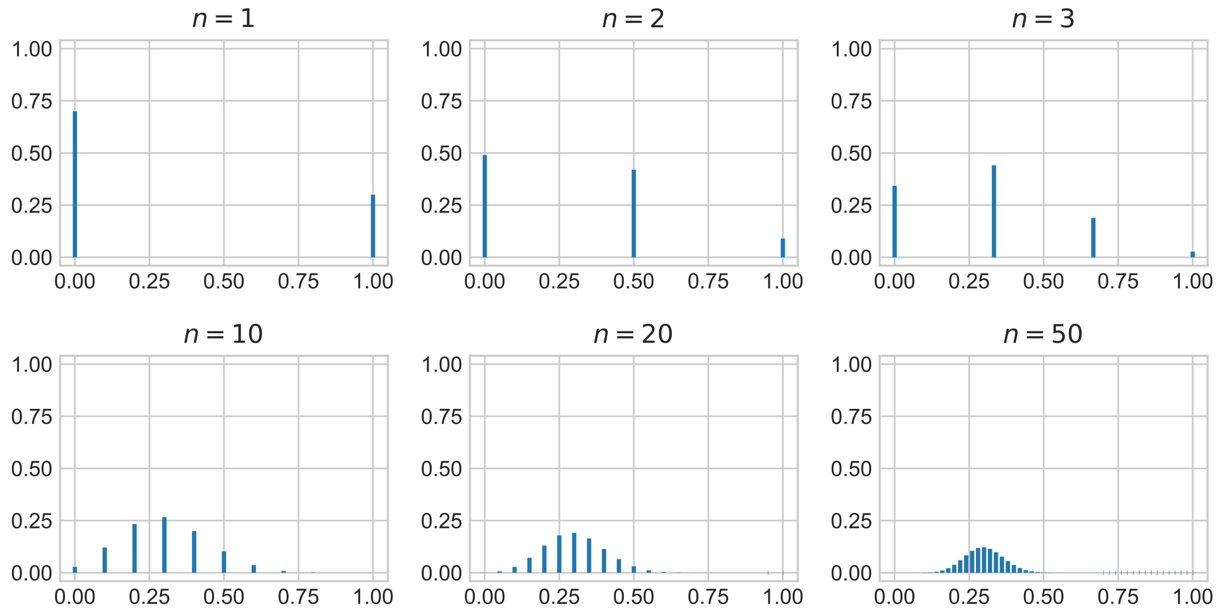

Figure 1: Probability mass function of $\overline{X}_n$ for $p = 0.3$.

Since $\mathbb{E}(\overline{X}_n) = p$, $\overline{X}_n$ is an unbiased estimator of p. Moreover, when n is large, the variance of $\overline{X}_n$ is small, so $\overline{X}_n$ is a good estimator of p.

Suppose that we have an unfair coin with $p = 0.3$. Figure 1 shows the probability mass functions of the sample average $\overline{X}_n$, where the number of coin tosses $n \in \{1, 2, 3, 10, 20, 50\}$. Since $p < 0.5$, we are more likely to see a smaller number of heads than that of tails in any given n tosses. In general, the probability mass function of $\overline{X}_n$ tends to skew towards to the right. However, as one can see in the plots in Fig. 1, the probability mass function becomes more and more symmetric as n gets bigger and bigger. This phenomenon arises for any $p \in (0, 1)$, no matter how extreme p is. Why is this happening? The answer is the Central Limit Theorem, which we have already learned in class.

The Central limit theorem tells us that $\overline{X}_n$ is approximately normally distributed for large n, i.e., $\overline{X}_n$'s distribution is approximately $\mathcal{N}(p, \frac{pq}{n})$.

Let us numerically verify it. In Fig. 2, for $p = 0.3$ and different values of n, we plot the probability density function of $\mathcal{N}(p, \frac{pq}{n})$ in dashed red, and we plot n times the probability mass function of $\overline{X}_n$ in blue. We can see that as n increases, the two curves match up quite well.

1.2 Exponential Random Variables

Let us now try applying the Central limit theorem to the exponential distribution. Let X_1, …, X_n be n independent random variables such that for all i from 1 to n, we have $X_i \sim \text{Exp}(\lambda)$, i.e., X_i is exponentially distributed with parameter λ. Recall that the probability

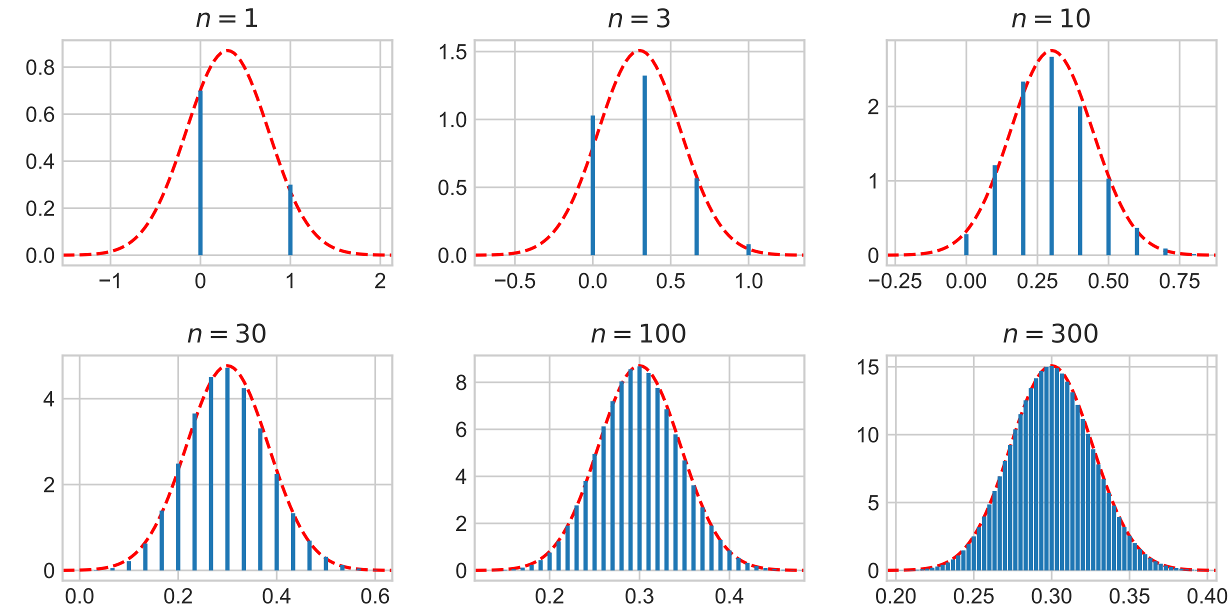

Figure 2: For $p = 0.3$, the blue bars are n times the probability mass function of $\overline{X}_n$, and the red curve is the probability density function of the normal distribution $\mathcal{N}(p, \frac{pq}{n})$.

density function of an exponential random variable with parameter λ is

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0,$$

the cumulative distribution function is

$$F(x) = 1 - e^{-\lambda x}, \quad x \geq 0,$$

the mean is $1/\lambda$, and the variance is $1/\lambda^2$.

Define S_n and $\overline{X}_n$ just like before:

$$S_n := \sum_{i=1}^{n} X_i, \qquad \overline{X}_n := \frac{1}{n} S_n.$$

Let us now find the mean and variance of $\overline{X}_n$. In Section 1.1, we knew the distribution of S_n. Here it's not so easy to see how S_n might be distributed. The following result can help us find $\overline{X}_n$ without first finding out the distribution of S_n.

Theorem 1.2. *Let $Y_1, \ldots, Y_n$ be n independent random variables. Let $Z := Y_1 + \ldots + Y_n$. Then*

$$\mathbb{E}(Z) = \sum_{i=1}^{n} \mathbb{E}(Y_i), \qquad \operatorname{Var}(Z) = \sum_{i=1}^{n} \operatorname{Var}(Y_i).$$

Exercise 1.1. Show that

(a) $\mathbb{E}(\overline{X}_n) = 1/\lambda$.

(b) $\mathrm{Var}(\overline{X}_n) = 1/(n\lambda^2)$.

The central limit theorem tells us that $\overline{X}_n$ is approximately normally distributed when n is large. By the results of Exercise 1.1, we get that $\overline{X}_n$ is distributed roughly as $\mathcal{N}(1/\lambda, 1/(n\lambda^2))$.

Let us try to verify the central limit theorem for $\overline{X}_n$ numerically.

Exercise 1.2. It can be shown that the probability density function of S_n is given by

$$f_{S_n}(x) = \frac{\lambda^n}{(n-1)!} x^{n-1} e^{-\lambda x}, \quad x \geq 0.$$

Define

$$Y_n := \frac{\overline{X}_n - \mathbb{E}(\overline{X}_n)}{\sqrt{\mathrm{Var}(\overline{X}_n)}} = \frac{\lambda S_n - n}{\sqrt{n}}.$$

Show that the probability density function of Y_n is

$$f_{Y_n}(y) = \frac{\sqrt{n}}{(n-1)!} \left(n + y\sqrt{n}\right)^{n-1} \exp\left(-(n + y\sqrt{n})\right), \quad y \geq -\sqrt{n}.$$

Hint: $F_{Y_n}(y) = \mathrm{P}(Y_n \leq y) = \mathrm{P}(S_n \leq (n + y\sqrt{n})/\lambda) = F_{S_n}((n + y\sqrt{n})/\lambda)$. Now differentiate both sides w.r.t. y.

By the central limit theorem, $\overline{X}_n$ should be distributed approximately $\mathcal{N}(\mathbb{E}(\overline{X}_n), \mathrm{Var}(\overline{X}_n))$. Thus, Y_n should be distributed approximately $\mathcal{N}(0,1)$. Since we found the true distribution of Y_n in Exercise 1.2, let us plot it against the standard normal distribution to see how well the central limit theorem holds.

Exercise 1.3. Use `matplotlib.pyplot` to plot all of following functions in the same figure:

1. For $n \in \{1, 3, 10, 30, 100\}$, plot $f_{Y_n}(y)$ for $y \in [-\min(4, \sqrt{n}), 4]$.
   (Y_n is defined in Exercise 1.2.)

2. Plot the standard normal distribution's density function on the interval $[-4, 4]$ as a dashed line. (You may use `scipy.stats.norm.pdf(x)` to get the standard normal distribution's PDF at x.)

Ensure your curves are properly labeled with a legend.

In general, we can conclude that Central Limit Theorem tells us that the distribution of an average tends to be Normal, even when the distribution from which the average is computed is non-Normal. However, there are certain cases in which the average deviates from Normal behavior. When does the Central Limit Theorem not hold?

1.3 Heavy-Tailed Distributions

Heavy-tailed distributions are probability distributions with a heavier tail than the exponential. We will see how the extremes produced by heavy-tailed distributions will corrupt the average so that an asymptotic behavior different from the Normal behavior is obtained. Formally, a random variable X is said to have a heavy-tail if

$$\lim_{x \to \infty} \frac{e^{-\lambda x}}{1 - F(x)} = 0, \qquad \text{for all } \lambda > 0.$$

Let us study the Pareto distribution. It has two positive parameters z and α, and its probability density function is given by

$$f(x) = \frac{\alpha z^{\alpha}}{x^{\alpha+1}}, \qquad x \geq z.$$

Its cumulative distribution function is

$$F(x) = 1 - \left(\frac{z}{x}\right)^{\alpha}, \qquad x \geq z.$$

The Pareto distribution is very important in reinsurance so we will study it closely.

Exercise 1.4. Show that

(a) A Pareto random variable is heavy-tailed. (Hint: L'Hôpital's rule.)

(b) An exponential random variable with parameter α is not heavy-tailed.

Exercise 1.5. Let $X \sim \text{Pareto}(z, \alpha)$.

(a) For what values of α does $\mathbb{E}(X)$ exist? Express $\mathbb{E}(X)$ in terms of α when $\mathbb{E}(X)$ exists.

(b) For what values of α do $\mathbb{E}(X)$ and $\text{Var}(X)$ exist? Express $\text{Var}(X)$ in terms of α when $\mathbb{E}(X)$ and $\text{Var}(X)$ exist.

Exercise 1.6. The Pareto distribution is closely related to the exponential distribution. Let $X \sim \text{Pareto}(z, \alpha)$ and $Y = \ln(X/z)$. Show that $Y \sim \text{Exp}(\alpha)$.
Hint: Use the same trick as in the hint of Exercise 1.2.

Since the Pareto distribution may not have a finite mean or variance, the Central limit theorem is not applicable.

2 QQ Plot

The quantile-quantile plot, also known as a QQ plot, is a tool to visually compare two distributions. It can also be used to compare a dataset of samples to a known distribution.

2.1 QQ Plots of Two Distributions

Definition 2.1 (Quantile function)**.** The *quantile function* of a distribution is, intuitively, the inverse of the cumulative distribution function. Formally, for any $p \in [0,1]$ and any random variable X, its quantile at p, denoted as $Q_X(p)$, is the largest value x such that $F_X(x) \leq p$. In other words, $Q_X(p) \geq x$ if and only if $F_X(x) \leq p$.

Example 2.1 (Exponential distribution)**.** Let $X \sim \text{Exp}(\lambda)$. Then for any $p \in [0,1]$,

$$F_X(x) \leq p \iff 1 - e^{-\lambda x} \leq p \iff e^{\lambda x} \leq \frac{1}{1-p} \iff x \leq \frac{1}{\lambda} \ln\left(\frac{1}{1-p}\right).$$

Thus,

$$Q_X(p) = \frac{1}{\lambda} \ln\left(\frac{1}{1-p}\right).$$

Let X and Y be two random variables. The QQ plot of Y w.r.t. X is obtained by plotting the points $(Q_X(p), Q_Y(p))$ for all $p \in [0,1]$.

QQ plots can help us compare two distributions. The idea is that if two distributions are similar, then their quantile functions will be similar too, and so, the plot will be very close to the line $y = x$.

QQ plots can also help us know if one random variable is a linear function of the other; in that case, the QQ plot will be a straight line because of the following result.

Theorem 2.1. *Let X be a random variable and $Y := aX+b$. Then $Q_Y(p) = aQ_X(p)+b$.*

Proof. For any number y, we have

$$F_Y(y) = \mathrm{P}(Y \leq y) = \mathrm{P}(X \leq (y-b)/a) = F_X((y-b)/a).$$

Hence, for any $p \in [0,1]$, we have

$$\begin{aligned} Q_Y(p) \geq y &\iff F_Y(y) \leq p \iff F_X((y-b)/a) \leq p \\ &\iff Q_X(p) \geq (y-b)/a \iff aQ_X(p) + b \geq y. \end{aligned}$$

Hence, $Q_Y(p) = aQ_X(p) + b$ for all $p \in [0,1]$. □

In Fig. 3 we have the QQ plot of the standard normal distribution w.r.t. the exponential distribution ($\lambda = 1$)

2.2 Tail Behavior from QQ Plot

The QQ plot can tell us whether two distributions are roughly the same, but it can help us in other ways too. By analyzing how the QQ plot differs from a straight line at the left and right ends of the plot, we can compare the tails of the two distributions.

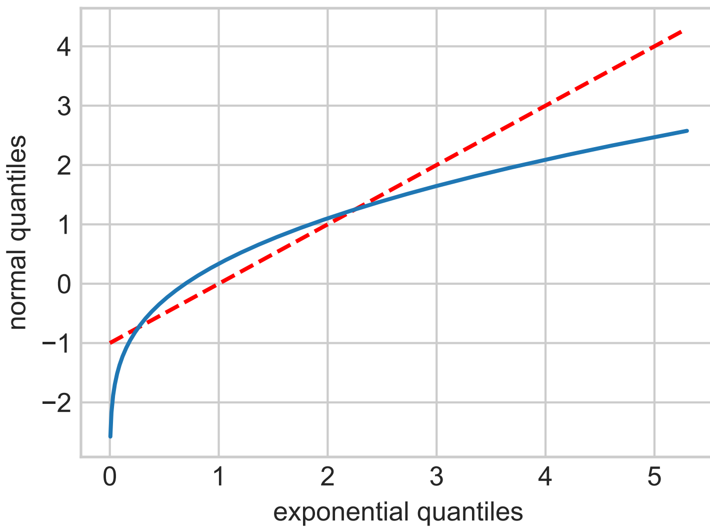

Figure 3: The blue line is the QQ plot of $Y \sim \mathcal{N}(0, 1)$ w.r.t. $X \sim \text{Exp}(1)$. The red dashed line is the equation $y = x - 1$.

To do that, we first draw the *reference line*. In a QQ plot of Y w.r.t. X, the reference line has the equation

$$y = \mathbb{E}(Y) + \sqrt{\frac{\text{Var}(Y)}{\text{Var}(X)}}(x - \mathbb{E}(X)).$$

1. When $Y = aX + b$, the reference line becomes $y = ax + b$.
2. For the example in Fig. 3, the reference line has the equation $y = x - 1$ and is drawn dashed red.

In Fig. 3, on the right side of the figure, the QQ plot curves downwards compared to the reference line. This means that as p increases and gets closer to 1, $Q_X(p)$ grows faster than $Q_Y(p)$. Intuitively, when a distribution has a heavy right tail, we see more extreme (larger) values, which suggests that Y has a lighter right tail than X. Indeed, X has PDF $f_X(x) = e^{-x}$, and Y has PDF $f_Y(x) \propto e^{-x^2/2}$. As x grows large, $e^{-x^2/2}$ becomes much smaller than e^{-x}, which tells us that Y has a lighter right tail than X. We can clearly compare the thickness of the tails in Fig. 4.

Let us now look at the left side of Fig. 3. The QQ plot curves downwards compared to the reference line. This means that as p decreases and gets closer to 0, $Q_Y(p)$ decreases much faster than $Q_X(p)$. Intuitively, when a distribution has a heavy left tail, we see more extreme (smaller) values, which suggests that Y has a heavier left tail than X. Indeed, for $x < 0$, the PDF of X is 0, and Y has PDF $f_Y(x) \propto e^{-x^2/2}$, so Y has a heavier left tail than X.

Since Y has a heavier left tail than X and a lighter right tail than Y, we get that Y is left skewed compared to X.

Let us summarize our observations.

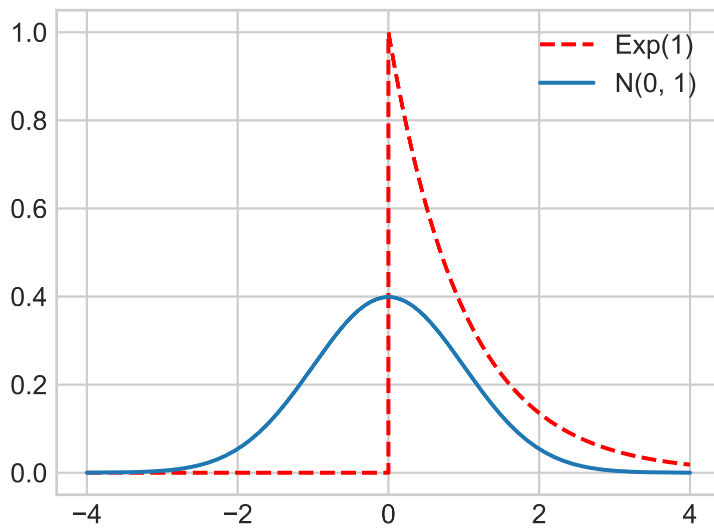

Figure 4: The blue curve is the PDF of $Y \sim \mathcal{N}(0,1)$. The red curve is the PDF of $X \sim \text{Exp}(1)$.

Let X and Y be any two random variables. In the QQ plot of Y w.r.t. X, compared to the reference line,

1. If the right end of the QQ plot
   (a) curves downwards, then Y has a lighter right tail than X.
   (b) curves upwards, then Y has a heavier right tail than X.
2. If the left end of the QQ plot
   (a) curves downwards, then Y has a heavier left tail than X.
   (b) curves upwards, then Y has a lighter left tail than X.

Exercise 2.1. Let $X \sim \mathcal{N}(0,1)$ have the standard normal distribution. Let $Y \sim U(-1,1)$ be uniformly distributed between -1 and 1.

(a) What is the quantile function of Y?

(b) What is the equation of the reference line?

(c) Use `matplotlib.pyplot` to draw the QQ plot of Y w.r.t. X. Also plot the reference line (dashed). Label the x and y axes of your plot (similar to Fig. 3). You can use `scipy.stats.norm.ppf(x)` as the quantile function of X.

(d) From the QQ plot, what can you say about the tails of Y compared to the standard normal distribution and why?

2.3 QQ Plots of Sampled Data

Often, we don't know the distribution of a random variable, but we can observe samples from it, and use them to get an *estimated QQ plot*.

Suppose we observe n samples from a distribution $\mathcal{D}$. After sorting them in non-decreasing order, let $(y_1, y_2, \ldots, y_n)$ be the resulting sequence. We define the *estimated quantile function* $\widehat{Q}$ as follows:

1. Let $p_i := \frac{i-1/2}{n}$ for all i from 1 to n.

2. Let $\widehat{Q}(p_i) = y_i$ for all i from 1 to n.

Then we draw the estimated QQ plot w.r.t. a random variable X as follows:

1. Let $x_i := Q_X(p_i)$ for all i from 1 to n.

2. Plot the points (x_i, y_i) for all i from 1 to n.

We demonstrate this process in Fig. 5. We sample 1000 points from the standard normal distribution, and draw the estimated QQ plot w.r.t. the exponential distribution with parameter 1.

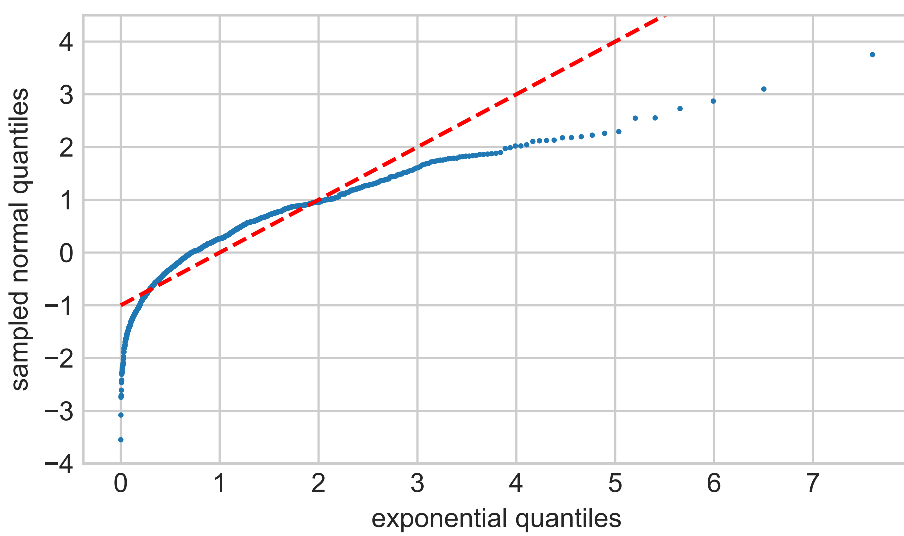

Figure 5: Estimated QQ plot of 1000 samples drawn from the standard normal distribution w.r.t. Exp(1). Observe the similarity with Fig. 3.

Let us try to see if the Pareto distribution follows the Central limit theorem. To do so, we sample 1000 points from Pareto$(1, 3/2)$ and compute the average. We repeat this process 1000 times to obtain 1000 sample averages. If the central limit theorem holds here, then these sample averages would have a normal distribution, and a QQ plot w.r.t. the standard normal distribution would give us a straight line. However, this does not happen, as seen in Fig. 6. Thus, we observe that the central limit theorem does not hold for Pareto$(1, 3/2)$.

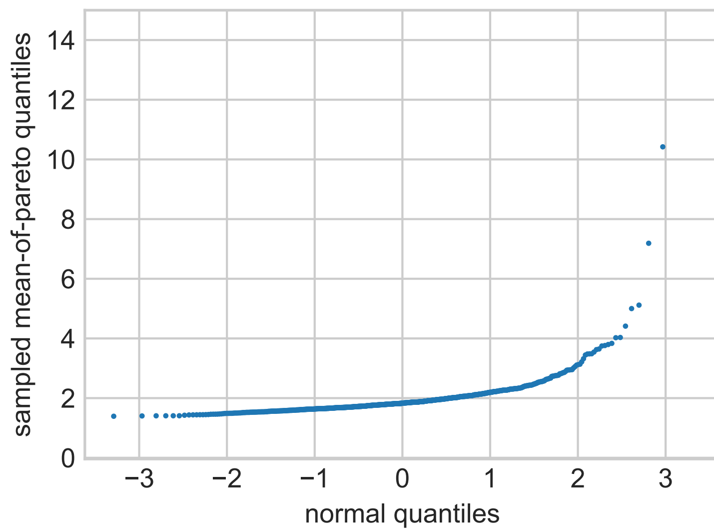

Figure 6: Estimated QQ plot of 1000 sample averages of 1000 points drawn from Pareto$(1, 3/2)$ w.r.t. $\mathcal{N}(0, 1)$. Note how the curve quickly starts jumping up as we move to the right. In fact, we had to remove the rightmost point $(3.2905, 259.78)$ from the plot to make it easier to see.

Exercise 2.2. Why does the central limit theorem not hold for Pareto$(1, 3/2)$? (Hint: recall the conditions for applying the central limit theorem.)

Let us create a QQ plot using real data.

The dataset `insurance.csv` contains insurance claim data from a reinsurance company. It contains 371 automobile claims from 1988 to 2001 that are greater than 1,200,000 euro. There are two columns: year and amount (in euros).

Exercise 2.3. We will create QQ plots using the `insurance.csv` data to learn about how the amounts are distributed.

1. Read `insurance.csv` and get the largest 270 amounts. Then apply the natural logarithm to each amount.

2. Create two QQ plots for this data (i.e., log of amounts), one w.r.t. the exponential distribution with parameter 1, and one w.r.t. the standard normal distribution. Draw as scatter plots. Label the x-axes appropriately. You are not required to draw the reference line.

3. Based on the QQ plots, which distribution does the data (i.e., log of amounts) fit better? Justify your answer.

3 Mean Excess Function and Reinsurance

Reinsurance is an insurance policy purchased by an insurance company from another insurance company, known as the 'reinsurer', as a means of risk management. It is a very common market practice when insurance companies undertake high risk profiles with potential catastrophic losses. A reinsurance agreement details the conditions upon which the reinsurer would pay a share of the claims incurred by the insurer and the reinsurer is paid a 'reinsurance premium' by the insurer, which issues insurance policies to its own policyholders.

A common form of reinsurance is the excess of loss (XL) reinsurance, where the insurer covers insurance claims from policyholders up to its *retention level* and any amount beyond the retention level will be reimbursed by the reinsurer. For example, an insurance company might insure commercial property risks with policy limits up to \$9 million, and then buy reinsurance of \$4 million on top of their retention level of \$5 million. In this case a loss of \$6 million on that policy will result in the recovery of \$1 million from the reinsurer.

3.1 Excess Function

The modeling of the XL reinsurance relies on an important mathematical concept called the *mean excess function*. Suppose a ceding insurer enters into an XL treaty with a retention level t. Let X be a random variable governing the size of a particular policyholder's claim. After claim investigation, the ceding insurer will make the payment, and the reinsurer has to pay $X - t$ if $X > t$. Pricing actuaries from reinsurance companies would want to know the average cost of such claims, which is theoretically determined by the mean excess function

$$e_X(t) := \mathbb{E}(X - t \mid X > t) = \mathbb{E}(X \mid X > t) - t. \tag{1}$$

Given the distribution of X, how can we find $e_X(t)$? The following result is helpful.

Theorem 3.1. *For any random variable X and a number t, we have*

$$e_X(t) = \frac{\mathbb{E}(X\mathbf{1}(X > t))}{\mathrm{P}(X > t)} - t.$$

Here $\mathbf{1}(A)$ is 1 if event A is true and 0 otherwise.

Proof sketch. Let $Z := \mathbf{1}(X > t)$. Then

$$\begin{aligned}\mathbb{E}(XZ) &= \mathbb{E}(XZ \mid X > t)\,\mathrm{P}(X > t) + \mathbb{E}(XZ \mid X \le t)\,\mathrm{P}(X \le t)\\ &= \mathbb{E}(X \mid X > t)\,\mathrm{P}(X > t) = (t + e_X(t))\,\mathrm{P}(X > t).\end{aligned}$$ □

Exercise 3.1. Prove that

(a) If $X \sim \mathrm{Exp}(\lambda)$, then for all $t \ge 0$, we have $e_X(t) = 1/\lambda$.
(Hint: you may want to use *integration by parts*.)

(b) If $X \sim \mathrm{Pareto}(z, \alpha)$ and $\alpha > 1$, then for all $t \ge z$, we have $e_X(t) = t/(\alpha - 1)$.

3.2 Non-Parametric Estimators of the Fair Net Premium

If the policyholder submits a claim of X, and the retention level is r, the amount that the reinsurer would pay is

$$\begin{cases} X - r & \text{if } X > r \\ 0 & \text{otherwise} \end{cases} \quad = \quad (X-r)\mathbf{1}(X > r) \quad = \quad \max(X - r, 0).$$

The fair net premium[1], denoted by $\Pi(r)$, is the expected value of the above quantity. Thus,

$$\begin{aligned} \Pi(r) &= \mathbb{E}(\max(X - r, 0)) = \mathbb{E}((X-r)\mathbf{1}(X > r)) && (2) \\ &= \mathbb{E}(X - r \mid X > r)\,\mathrm{P}(X > r) = e_X(r)\overline{F}_X(r). && (3) \end{aligned}$$

Here $\overline{F}_X(r) := 1 - F_X(r) = \mathrm{P}(X > r)$ is called the *survival function*.

Let $x_1, x_2, \ldots, x_n$ be the claim amounts. Assume that they are sorted in non-increasing order, i.e., $x_1 \geq x_2 \geq \ldots \geq x_n$. Let us try to estimate $\Pi(r)$ using these claim amounts.

Our first estimator is based on Eq. (2):

$$\widehat{\Pi}_1(r) := \frac{1}{n}\sum_{i=1}^{n} \max(x_i - r, 0). \tag{4}$$

Since reinsurance contracts are meant to transfer risks of catastrophic losses, claims of small and medium sizes provide no useful information for the valuation of reinsurance contracts. For example, a retention level of $r = 5,000,000$ euros is typical in practice, but only 12 observations from `insurance.csv` are above this limit. Hence, the *effective sample size* for computing $\widehat{\Pi}_1(r)$ is just 12. Thus, $\widehat{\Pi}_1(r)$ is not a good estimator in this example.

Another way we can estimate $\Pi(r)$ is based on Eq. (3): we estimate $e_X(r)$ and $\overline{F}_X(r)$ separately, and the multiply them.

1. We estimate $e_X(r)$ using Eq. (1). We collect all claims above the retention level, take their mean, and then subtract the retention level. Call this estimator $\widehat{e}(r)$. Formally, $\widehat{e}(r) := \mathrm{sum}(S)/|S| - r$, where $S := \{x_i : x_i > r,\ i \text{ from } 1 \text{ to } n\}$.
2. $\overline{F}_X(r) := \mathrm{P}(X > r)$, so we estimate it as the proportion of claims in the set $\{x_1, \ldots, x_n\}$ above the retention level. Call this estimator $\widehat{\overline{F}}(r)$.

Finally, we define our new estimator of $\Pi(r)$ as $\widehat{\Pi}_2(r) := \widehat{e}(r) \cdot \widehat{\overline{F}}(r)$.

Unfortunately, it turns out that $\widehat{\Pi}_2(r)$ isn't a good estimator either, as we will see in the following exercise.

Exercise 3.2. Show (using algebraic manipulation) that $\widehat{\Pi}_1(r) = \widehat{\Pi}_2(r)$, regardless of which claims dataset $\{x_1, \ldots, x_n\}$ we have.

[1] Net premium refers to the pure cost of insurance coverage. It is used in contrast with gross premium, which includes commission, policy expenses, and other administrative costs.

3.3 Parametric Estimator of the Fair Net Premium

$\widehat{\Pi}_1$ was a bad estimator because we were essentially throwing away almost all data. Even though small claims are not relevant to reinsurance, they can help us understand the distribution of large claims.

Pareto distribution often provides a good fit for the distribution of large claims. Indeed, by combining the results of Exercises 1.6 and 2.3, we can infer that `insurance.csv` follows a Pareto distribution. What remains is estimating the parameters z and α of this distribution.

A well-known estimator of $1/\alpha$ was proposed by Hill [1]. The intuition behind this estimator follows from Exercises 1.6 and 2.3, that the mean excess function of the logarithm of a Pareto random variable is $1/\alpha$. Let $x_1, \ldots, x_n$ be samples from Pareto(z, α) such that $x_1 \geq \ldots \geq x_n$. The Hill estimator for $1/\alpha$ (parametrized by $k \in \{1, \ldots, n-1\}$) is

$$\widehat{\gamma}_k := \frac{1}{k} \sum_{j=1}^{k} \ln\left(\frac{x_j}{x_{k+1}}\right).$$

The Hill estimator enjoys a high degree of popularity thanks to some nice theoretical properties, which we do not intend to discuss in this class. For example, it has been proven that the Hill estimator is a consistent estimator for $1/\alpha$. One should note that the Hill estimator is based on the $k+1$ largest observations, and for each choice of k there would be an estimate of $1/\alpha$. Many researchers have developed strategies to determine the optimal k under various statistical criteria. The discussion of such strategies can be quite involved and hence is omitted here.

Exercise 3.3. Suppose we pick $k = 95$ for `insurance.csv`. Compute (and print) the Hill estimator $\widehat{\gamma}_k$.

To estimate the fair net premium $\Pi(r)$, we will use Eq. (3), i.e., we will estimate $e_X(r)$ and $\overline{F}_X(r)$ separately, and then multiply them to get an estimate of $\Pi(r)$. By Exercise 3.1, we get that

$$e_X(r) = \frac{r}{\alpha - 1}.$$

Estimating $\overline{F}_X(r) := \mathrm{P}(X > r)$ is not as straightforward. We could try estimating it like in Section 3.2, i.e., the proportion of claim amounts more than r, but we have already seen why this would be a bad idea: when r is large, very few observations would count, so the estimator would be very noisy. However, if r was small, this might have worked well. The trick here is to write

$$\mathrm{P}(X > r) = \frac{\mathrm{P}(X > r)}{\mathrm{P}(X > t)} \cdot \mathrm{P}(X > t).$$

Then we can estimate $\mathrm{P}(X > t)$ as the fraction of claim amounts more than t. Since $X \sim$ Pareto(z, α), we get

$$\frac{\mathrm{P}(X > r)}{\mathrm{P}(X > t)} = \frac{(z/r)^\alpha}{(z/t)^\alpha} = \left(\frac{t}{r}\right)^\alpha.$$

A good choice of t is x_{k+1}.

Finally, on combining all these observations, we get the following estimator for the fair net premium:

$$\widehat{\Pi}_3(r) = \frac{r}{1/\widehat{\gamma}_k - 1} \cdot \left(\frac{x_{k+1}}{r}\right)^{1/\widehat{\gamma}_k} \cdot \left(\frac{k}{n}\right) \tag{5}$$

Exercise 3.4. For `insurance.csv`, compute $\widehat{\Pi}_1(r)$ and $\widehat{\Pi}_3(r)$ for the following values of r: 3×10^6, 3.5×10^6, 4×10^6, 5×10^6, 7.5×10^6. Set $k = 95$ and use the Hill estimator from Exercise 3.3. Print your answers with two decimal places, i.e., round the premiums to the nearest cent.

Now you have completed a pricing exercise that an actuary would typically do for a reinsurance contract on luxury commodities. You can see that there is no 'one-size-fits-all' magic formula for pricing a product. Keep in mind that each of the estimators we developed has its own merits and limitations. Observe from your solution to Exercise 3.4 that the estimates of net premiums are close for small retention levels, as more data are used in all estimators. But they are far apart for high retention levels, due to the 'low quality' results from the non-parametric statistics with limited sample data. An advanced knowledge of heavy-tailed distribution allows us to make better use of data and provide more trust-worthy solutions in this example, which would be very important for billion-dollar businesses like reinsurance.

References

[1] Bruce M. Hill. A simple general approach to inference about the tail of a distribution. *The Annals of Statistics*, pages 1163–1174, 1975. doi:10.1214/aos/1176343247.

A Grading Rubric (50 points)

A.1 Central Limit Theorem (18 points)

1. Exercise 1.1 (2 points): 1 point for $\mathbb{E}(\overline{X}_n)$, 1 point for $\mathrm{Var}(\overline{X}_n)$.
2. Exercise 1.2 (2 points)
3. Exercise 1.3 (4 points): 1 point to plot the normal curve, 2 points to plot f_{Y_n}, 1 point for plot legend.
4. Exercise 1.4 (3 points): 2 points for (a), 1 point for (b).
5. Exercise 1.5 (5 points): 2 points for (a), 3 points for (b).
6. Exercise 1.6 (2 points)

A.2 QQ Plot (16 points)

1. Exercise 2.1 (7 points):
   - 1 point for (a).
   - 1 point for (b).
   - 4 points for (c): 2 points for the QQ plot, 1 point for the reference line, 1 point for plot labels.
   - 1 point for (d).
2. Exercise 2.2 (1 point)
3. Exercise 2.3 (8 points):
   - 2 points to read and transform the data.
   - 5 points to plot: 3 points to compute the (x, y) points, and for each of the two plots, 0.5 point to plot and 0.5 point to label the x-axis.
   - 1 point to identify the better-fitting distribution.

A.3 Mean Excess Function and Reinsurance (16 points)

1. Exercise 3.1 (6 points): 3 points for (a), 3 points for (b).
2. Exercise 3.2 (3 points):
3. Exercise 3.3 (4 points): 2 points to read and transform the data, 2 points to compute and print the hill estimator.
4. Exercise 3.4 (3 points): 1 point to compute $\widehat{\Pi}_1(r)$, 1.5 points to compute $\widehat{\Pi}_3(r)$, 0.5 points to print the outputs to 2 decimal places.